

Representational Geometry in Neural Networks

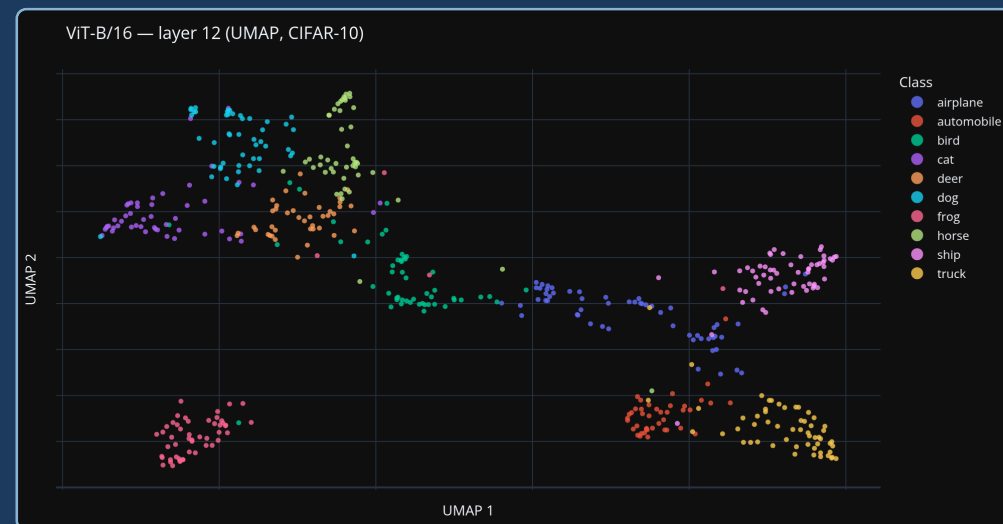
From Vision Transformers to RLHF-Aligned Language Models

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ViT-B/16 · layer 12 · CIFAR-10 · real data

Part 1: Cross-modal alignment

When a vision model and a language model train on the **same task**, does a shared representational structure emerge across modalities?

- **Hypothesis A:** cross-modal signal is new information → improves model
- **Hypothesis B:** models converge regardless; extra signal adds little

CKA before/after cross-modal fine-tuning · DINOv2
+ Llama · MS-COCO

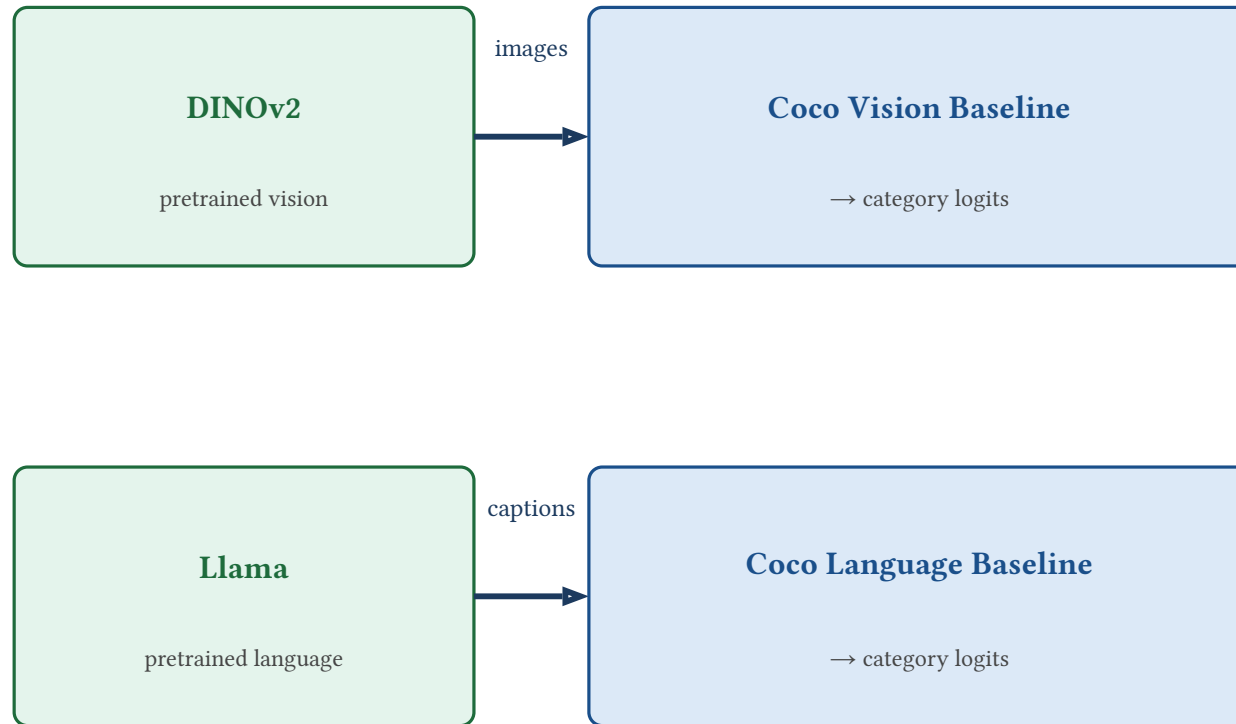
Part 2: RLHF geometry

Does RLHF geometrically transform a language model's embedding space, encoding human preference as a **linearly separable structure**?

- Does alignment progressively separate chosen/rejected responses across SFT → DPO → RLHF?
- How does separation evolve **layer by layer**?

UMAP + LinearSVC per layer · Tulu-3-8B SFT/DPO/
RLHF · Anthropic/hh-rlhf

Part 1 · Setup: Two Independent Baselines on MS-COCO



Part 1 · Example: MS-COCO 2017

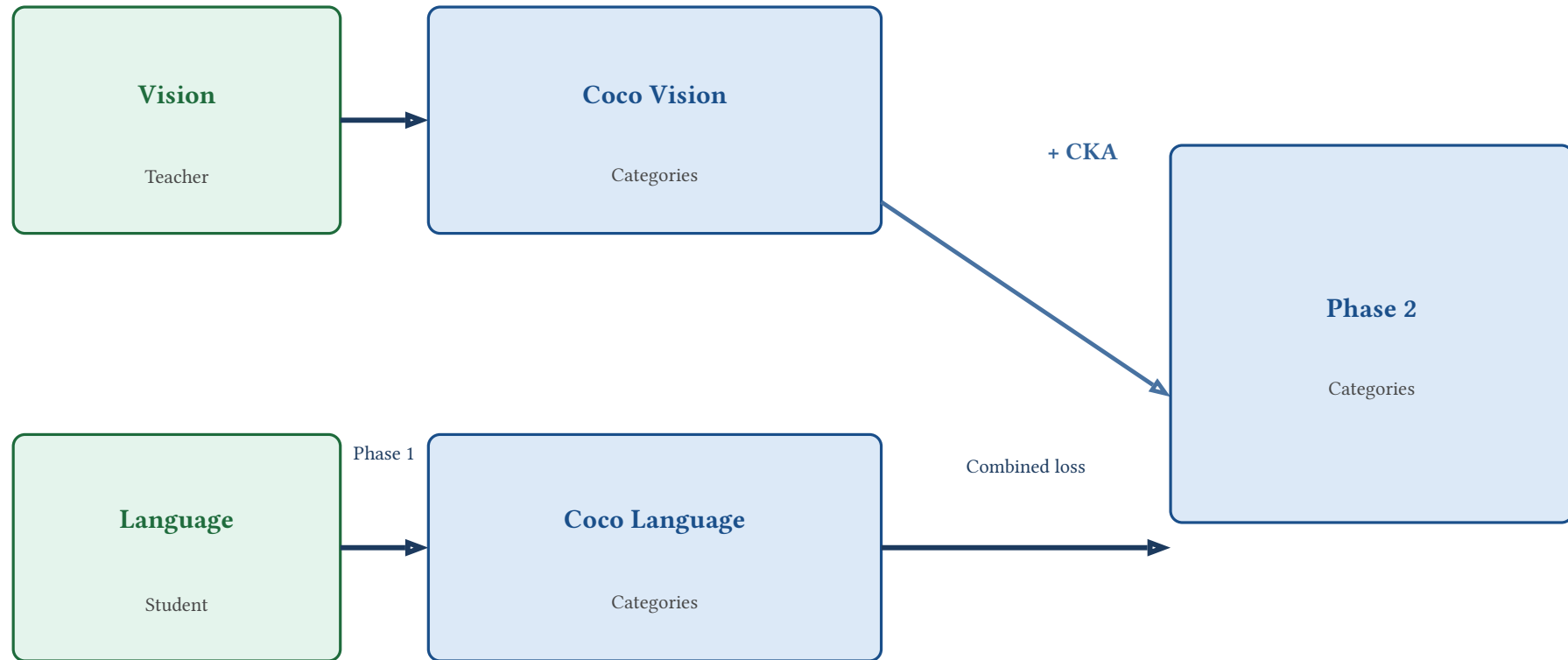
Categories: person, tie, umbrella, handbag

Captions:

- A girl in grey jacket and tie standing on a street.
- Little girl posing for the camera with an adult sized striped tie on.
- A kid with a tie standing in the street.
- A young girl wearing a necktie and smiling broadly.
- Girl smiles for picture in busy Asian plaza.



Part 1 · Method: Teacher → Student CKA Fine-tuning



Part 1: Cross-modal Alignment · Real Results

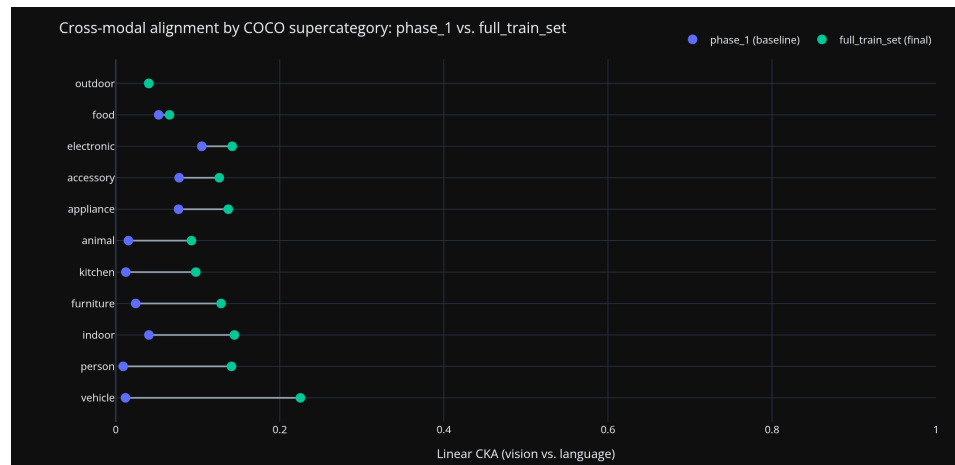
Alignment is built by training, not inherited

Baseline (phase_1): near-zero everywhere, mean CKA 0.005, max 0.007. No depth structure at all.

After CKA fine-tuning: mean 0.048, max 0.176, roughly **25×** higher

Peak stays at the **same** layer pair both times (vision layer 16 × language layer 28); training **amplifies** an existing affinity, doesn't create one elsewhere

Contradicts naive Platonic Representation Hypothesis reading: untrained models do **not** already converge here



Real dashboard export · CKA per COCO supercategory, phase_1 vs. full_train_set

By COCO supercategory

- 11/12 supercategories gained alignment; outdoor flat, none regressed
- Biggest movers: **vehicle** (+0.21), **person** (+0.13)

Part 2 · Motivation: Why RLHF?

Pretraining and SFT optimize **likelihood**, not **preference**.

Without RLHF

- Next-token prediction rewards **plausible** text
- No signal for helpful / harmless / honest
- A great answer and a subtly wrong one can look equally likely

With RLHF

- Optimizes directly for **human preference**
- Signal comes from comparisons, not a fixed label set
- Model improves on axes humans care about, not just fluency

Insight: Goal Shift probability mass toward responses humans rank higher, without collapsing fluency or drifting into reward hacking.

Classic formulation (Ouyang et al., 2022 · InstructGPT): the paradigm Part 2 measures the geometric effect of

1. Collect preference data 2. Train a reward model 3. Policy optimization

- Sample **multiple** completions per prompt from the SFT model
- Humans compare pairs $\rightarrow$ chosen y_w , rejected y_l
- Output: dataset of (x, y_w, y_l) triples

- $r_\varphi(x, y)$ learns to **score** a completion
- Bradley–Terry pairwise loss:
$$\mathcal{L}(\varphi) = -\mathbb{E}[\log \sigma(r_\varphi(x, y_w) - r_\varphi(x, y_l))]$$
- Learns to rank, not to generate

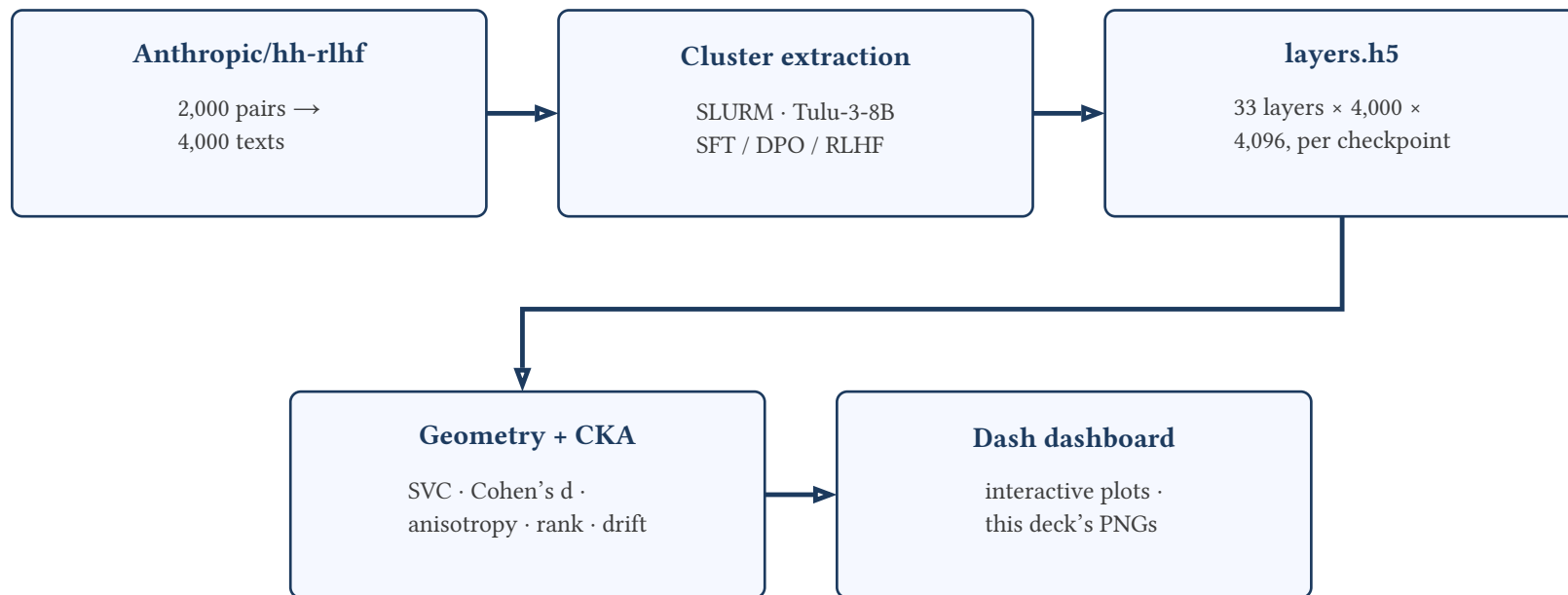
- π_θ generates, r_φ scores, PPO updates θ

$$\max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [r_\varphi(x, y)] - \beta \text{KL}(\pi_\theta \parallel \pi_{\text{ref}})$$

- KL term stops **reward hacking**

Remark Neither of our checkpoints follows this exactly: DPO skips the reward model entirely, and our “RLHF” checkpoint is actually **RLVR** (rule-based, verifiable rewards, no learned reward model), kept as “RLHF” only for consistency with Part 1’s vocabulary.

Part 2 · Pipeline: From Dataset to Dashboard



`scripts/extract_embeddings.py` → `app/rlhf_geometry_compute.py` /
`app/vision_compute.py` → `app/app.py`

Part 2 · What Each Checkpoint Actually Is

Tulu-3-8B alignment trajectory: 3 public AllenAI checkpoints, same base model, extracted at each post-training stage

SFT

Llama-3.1-Tulu-3-8B-SFT

- Instruction fine-tuning
- No preference signal yet

DPO

Llama-3.1-Tulu-3-8B-DPO

- Chosen > rejected, directly
- No reward model

RLHF

Llama-3.1-Tulu-3-8B

- Final RL stage (Tulu-3's RLVR)
- Labeled **RLHF** for Part 1 consistency

Remark All 3 are the **same** 8B model at successive training stages, not different architectures: a clean trajectory, not a cross-model comparison.

Tulu-3-8B SFT → DPO → RLHF · Anthropic/hh-rlhf · 4,000 samples · 33 layers

Negative finding

- Preference not linearly encoded, any stage
- LinearSVC peak: 0.543 / 0.544 / 0.550, near chance
- No phase transition in preference geometry

But the geometry does shift

- CKA(SFT, DPO) drops to **0.976** by layer 32
- CKA(DPO, RLHF) stays near **0.999**
- Most of the shift happens SFT → DPO

No readable linear “preference direction”

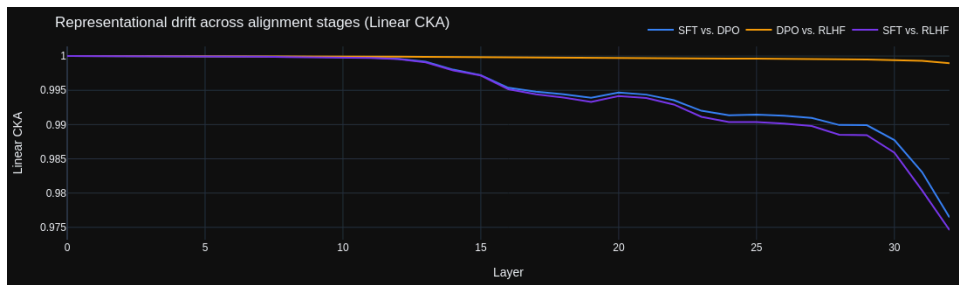
- Flat, near-chance at every layer, all 3 checkpoints
- DPO/PPO-style objectives preserve or degrade separability, not universal to “RLHF”

But geometry is not static

- Real reorganization, concentrated at **SFT** → **DPO**
- Most “alignment work” happens before PPO

*Alignment reshapes geometry **globally**, not a single axis.*

Deliverable & Honest Limitations



Interactive Plotly Dash · CKA heatmaps · layer sliders · UMAP/t-SNE · SVC score

Project deliverable

The dashboard **is** the deliverable, an interactive visualization of all findings: layer slider, CKA heatmaps, UMAP scatter, SVC score

Limitations & learnings

- **Part 1:** float16 overflow in extraction; later vision layers, most phase_1 language layers came back inf
- **Part 2:** last-token pooling only; paired-difference probes reach 84% in the literature vs. our 55% peak

He, Trott, Khosla (2025)

Shared Latent Representations across Vision and Language

arXiv:2509.20751 · **Anchor paper**

Kornblith et al. (2019)

Similarity of Neural Network Representations Revisited

ICML · arXiv:1905.00414 · **CKA method**

Christiano et al. (2017)

Deep RL from Human Preferences

NeurIPS · arXiv:1706.03741 · **RLHF foundations**

Lambert et al. (2024)

Tulu 3: Pushing Frontiers in Open Language Model Post-Training

arXiv:2411.15124 · **Models used for Part 2**

Kucukahmetler et al. (2026)

Relative Geometry of Neural Forecasters

TMLR · arXiv:2602.15676

Ouyang et al. (2022)

Training LMs to Follow Instructions with Human Feedback

NeurIPS · arXiv:2203.02155 · **InstructGPT / RLHF**

McInnes et al. (2018)

UMAP: Uniform Manifold Approximation and Projection

arXiv:1802.03426 · **Dim. reduction method**

Huh, Cheung, Wang, Isola (2024)

Position: The Platonic Representation Hypothesis

ICML · arXiv:2405.07987 · **“Models converge inherently” framing**

References: Part 2 (RLHF Geometry)

Bai et al. (2022)

Training a Helpful and Harmless Assistant with RLHF
arXiv:2204.05862 · **Original hh-rlhf dataset paper**

Sinha, Garg, Elluru, Singh, Garg (2026)

Mechanistic Analysis of Alignment Algorithms in Language Models
arXiv:2606.09850 · **Objective-dependent separability · re-check**
Stiennon et al. (2020)

Learning to Summarize from Human Feedback
NeurIPS · arXiv:2009.01325 · **77%±2% agreement, contrast to hh-rlhf**
Zhao, Gong, Chen, Kang, Li (2026)
A Layer-wise Analysis of Supervised Fine-Tuning
arXiv:2604.11838 · **Final layers most SFT-sensitive · re-check**

Lee, Bai, Pres, Wattenberg, Kummerfeld, Mihalcea (2024)

A Mechanistic Understanding of Alignment Algorithms: A Case Study on DPO and Toxicity
ICML · arXiv:2401.01967 · **DPO learns an “offset”, not erasure**

Ziegler et al. (2019)

Fine-Tuning Language Models from Human Preferences
arXiv:1909.08593 · **Labeler agreement baselines**
Kirin (2026)

Relational Preference Encoding in Looped Transformer Internal States
arXiv:2604.09870 · **Paired-diff probes reach 84.5% · re-check**
Kumar et al. (2026)
When Synthetic Speech Is All You Have: Better Call GRPO
arXiv:2607.08409 · **GRPO changes behavior, not geometry · re-check**